



Technische Hochschule
Ingolstadt

Bachelor Thesis

Faculty of computer sciencethesis.tex

**Development of a Frontend Application for Real-Time and
24-Hour Monitoring and Recording of Electrocardiographic Data
with Communication via Bluetooth Low Energy**

First- and Lastname : **Eric Horacsek**

Registered on : xx.yy.zzzz

Submitted on : xx.yy.zzzz

First examiner : Prof. Dr.-Ing. Georg Passig

Second examiner : Prof. Dr. Ulrich Margull

AFFIDAVIT

I hereby declare that I have written this thesis independently, have not used any sources or aids other than those stated, have not submitted it elsewhere for examination purposes and have expressly identified all quotations from the sources used, either verbatim or in spirit. I have marked verbatim and indirect quotations as such.

Ingolstadt, _____

Firstname Lastname

ACKNOWLEDGEMENTS

This is the sentimental part where you get to thank all the persons who were a part of your thesis journey in one or the other way!

Eric Horacsek
Ingolstadt, Deutschland
November xx 2024

ABSTRACT

The human heart is the central organ of the circulatory system and plays a vital role in supplying the body with oxygen and nutrients. Rhythmic contractions pump blood through the body, laying the foundation for all vital functions. An essential part of medical diagnostics combined with heart function and health is the electrocardiogram (ECG). The ECG measures the electrical activity of the heart and provides valuable information about the heart rhythm as well as possible irregularities or diseases. As technology has advanced, the way in which ECG data can be collected and monitored has also evolved significantly. Mobile health monitoring in particular now makes it possible to record ECG data not only at certain points of time, but also in real time and over longer periods of time. Wireless transmission via Bluetooth Low Energy (BLE) enables reliable and energy-efficient transfer of data to mobile devices. This opens up new possibilities for applications offering real-time and long term monitoring enabling users access to relevant information and graphical processing and alarm functions. This thesis explains the conception and implementation of an Android frontend application, which realises continuous and mobile ECG monitoring. The main programming language used for this project, is kotlin with Jetpack Compose as the Framework. Moreover this dissertation is particularly focused on implementing the communication and data transmission via BLE as well as realising an 24-hour-mode, which might be used for 24-hour data recognition. It also provides an broad insight into the given requirements, designprincipals, and technical challenges which accured during the implementation. Furthermore this theseis offers a glimpse into the potential for modern healthsuveillance.

CONFIDENTIALITYCLAUSE

Optional.

Ingolstadt, _____
(Date)

(Signature)
Firstname Lastname

GLOSSARY

Eric Horacsek Absolute operation.

ACRONYMS

D Dimensions.

RGB Red Green Blue channels.

LIST OF FIGURES

LIST OF TABLES

LIST OF CONTENTS

Affidavit	I
Acknowledgements	II
Abstract	III
Confidentialityclause	IV
Glossary	V
Acronyms	V
List of figures	VI
List of tables	VII
1 Introduction	1
1.1 Motivation	1
1.2 Scope and Framework	1
1.3 Procedure	1
2 Technical Fundamentals	2
2.1 Selected Technologies	2
2.1.1 Programming Language and Framework	2
2.1.2 Relevanz für die eigene Arbeit - Direction 1	2
2.2 Research direction 2	2
2.2.1 Relevanz für die eigene Arbeit - Direction 2	3
2.3 Schlussfolgerung	3
3 Diskussion	4
4 Conclusion	5
4.1 Outlook	5
4.2 Einschränkungen	5
Literature references	VI

1 INTRODUCTION

This chapter will give a small introduction on what the thesis and the implemented application is about. This covers an explanation for the application's motivation and the environment the application will be used in.

1.1 Motivation

The aim of this work is to create a new version of an Android App once developed by fellow students from the faculty of Electrical and Information Technology. It's core functionality is to receive and display ECG data in real time, via BLE. With the completion of this work, the new version will be able to receive and display ECG data in real time, via BLE. In addition it will be capable of an 24-hour mode which records ECG data for up to 24 hours, so the collected data can be analyzed later.

1.2 Scope and Framework

This paper only deals with the app development process of the new version. Therefore, any implementation of the previous work done by the students is not going to be discussed in detail. The hardware implementation is also not an element of this work . Generally speaking, this paper covers all topics that are essential for the development of the new app version. Specifically, technology stack of choice, IDE choice desicion, communication with hardware via Bluetooth low energy, structure of the user interface, real-time data monitoring, implementing the 24 hour mode. In addition, the outlook explains procedures for the implementation of future features that could not be realized within the limits of this paper.

1.3 Procedure

2 TECHNICAL FUNDAMENTALS

"The nature of the application and the required functionality of a software system should influence the choice of methods and tools used during development". [Clarke \[1995\]](#) This ensures that the development process is both efficient and tailored to meet the unique demands of the system. Moreover, aligning tools and methods with the application's requirements can help streamline workflows and reduce potential complexities during implementation. Consequently choosing the right programming language, aswell as IDE is essential to enable efficient development.

The following chapter Technical Fundamentals contains all technical characteristics and requirements, and how they have been fulfilled.

2.1 Selected Technologies

2.1.1 Programming Language and Framework

When searching for a suitable programming language to program Android apps, names like Java/Kotlin and Dart often appear. Both languages offer great frameworks, such as Flutter for Dart or Jetpack Compose for Kotlin. One aswell as the other are highly effective because they eliminate the limitations of XML-based UI development, enabling a more streamlined, declarative approach that combines UI and logic in a single language. Thus reduces boilerplate code, and enhances performance aswell as developer productivity. The main difference between both frameworks lies in resource management. For example Flutter is superior in cpu usage, but Jetpack Compose needs less memory. [Kusuma et al. \[2023\]](#) "But since both have their respective advantages and disadvantages, it is impossible to say that one is superior to the other. The choice between Jetpack Compose and Flutter depends on the needs and preferences of the developer. If developers are more familiar with the Kotlin language and want to use new technologies, then Jetpack Compose might be a better choice." [Kusuma et al. \[2023\]](#) With prior practical experience in Java/Kotlin, Jetpack Compose was chosen as the framework for this work.

2.1.2 Development Environment

Lorem ipsum Lorem ipsum.

2.2 Research direction 2

[Wintersberger et al. \[2017\]](#) befasst sich mit einer umfassenden und klar definierten Taxonomie der zwei Hautprozesse "handover" und "handback". Er kommt zu folgender Schlussfolgerung:

1. Handover - von der automatisierten zur manuellen Fahrt
2. Handback - von der manuellen zur automatisierten Fahrt

2.2.1 Relevanz für die eigene Arbeit - Direction 2

Lorem ipsum Lorem ipsum.

2.3 Schlussfolgerung

Zusammenfassung des Kapitels, Ableitung von Empfehlungen, etc. für die eigene Arbeit.

- Lorem ipsum subsection 2.1.
- Lorem ipsum lorem ipsum subsection 2.1.
- Lorem ipsum subsection 2.2.
- ...
- Lorem ipsum lorem ipsum lorem ipsum subsection 2.2.

3 DISKUSSION

Diskussion der Resultate im Vergleich zu Ergebnissen der Related work-Analyse und Reflektion auf die Forschungsfragen/Hypothesen.

Hypothese **H1** kann angenommen werden. Bla bla Begründung/Argumentation.

Andererseits zeigen die Ergebnisse der Evaluation aber keinen signifikanten Unterschied (siehe **Field**) zwischen X und Y - somit kann auch **H2** angenommen werden. Da hier bla bla Argumentation.

Durch die Ergebnisse des Fragebogens wird Hypothese **H3** nicht bestätigt. Es sind keine signifikanten Unterschiede über die ...

4 CONCLUSION

Diese Arbeit befasste sich grundsätzlich mit der Frage, welche Anforderungen an ... gestellt werden. Der Lösungsvorschlag war... Benutzerstudien haben gezeigt, dass es einen signifikanten Unterschied... gibt. Diese Ergebnisse motivieren, um ...

4.1 Outlook

In dieser Arbeit haben wir uns mit ... beschäftigt. Durch zwei Benutzerstudien wurde festgestellt, dass sich die Aufteilung und Positionierung der Informationen innerhalb der Anzeige der beiden Gruppen aufgrund bla bla ändert. Das entwickelte Konzept ist zwar bla bla, müsste aber für eine Wiederholung der Studien bla bla zur Erhebung quantitativer Daten wie folgt angepasst werden:

- xxx
- yyy
- xzz

Des Weiteren wurde festgestellt, dass bla bla. Auch das müssten zukünftige Varianten besser berücksichtigen, z. B., indem sie bla bla. Für Studien sollte zudem eine neutrale(re) Umgebung gewählt werden. Somit sollte sichergestellt werden können, dass beispielsweise ein möglicher Bias der Marke des Fahrzeugs sich nicht auf das HMI-Konzept auswirkt...

4.2 Einschränkungen

Hinsichtlich Einschränkungen, die eine breite Anwendung der Ergebnisse verhindern, sind zwei getrennte Bereiche zu betrachten. Zum einen die Evaluation, welche sich nur auf den ersten Teil der Arbeit bezieht, zum anderen das ausgearbeitete Konzept für die Anzeige, das auf dem ersten Teil der Arbeit aufbaut.

Evaluation Bla bla.

Konzept Bla bla. Das wurde etwa in beschrieben...

LITERATURE REFERENCES

- S.J. Clarke. Selecting the correct tools and methods for software systems development. In *IEE Colloquium on Are Software Development Technologies Delivering Their Promise?*, page 7/1, 1995. doi: 10.1049/ic:19950387.
- Andy Field. *Discovering Statistics Using SPSS*. SAGE Publications Ltd, 3 edition. ISBN 978-84787-906-6.
- Mandahadi Kusuma, Ahmad Haris Rifani, and Bambang Sugiantoro. Comparison analysis of jet-pack compose and flutter in android-based application development using technical domain. In *2023 Eighth International Conference on Informatics and Computing (ICIC)*, pages 3–5, 2023. doi: 10.1109/ICIC60109.2023.10381987.
- Philipp Wintersberger, Paul Green, and Andreas Riener. Am i driving or are you or are we both? a taxonomy for handover and handback in automated driving. In *Proceedings of the Ninth International Driving Symposium on Human Factors in Driver Assessment, Training and Vehicle Design*, Driving Assessment Conference. University of Iowa, June 2017. URL <http://drivingassessment.uiowa.edu/sites/default/files/DA2017/papers/51.pdf>.